

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 12

Regulation of Carbohydrate Metabolism

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Objectives: Regulation of carbohydrate metabolism

SOM. MK.I.BPM2.2.DM.2.BCHM.1144	Discuss the concept of reciprocal regulation of glycolysis and gluconeogenesis in the fed state and during fasting.
SOM. MK.I.BPM2.2.DM.2.BCHM.1145	Explain the use of glycogen as a source of glucose in skeletal muscle and outline regulation of glycolysis in muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1146	Analyze the role of calcium and AMP in regulation of glycogenolysis in skeletal muscle
SOM. MK.I.BPM2.2.DM.2.BCHM.1147	Describe the effect of epinephrine on muscle contraction.
SOM. MK.I.BPM2.2.DM.2.BCHM.1148	Describe metabolic changes in the muscle during exercise and describe ATP formation in muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1149	Review regulation of glucokinase and allosteric regulation of hepatic glycolysis (PFK-1 and feed-forward activation of pyruvate kinase).
SOM. MK.I.BPM2.2.DM.2.BCHM.1150	Describe the role of fructose 2,6-bisphosphate in reciprocal regulation of glycolysis and gluconeogenesis in the liver.
SOM. MK.I.BPM2.2.DM.2.BCHM.1152	Assess regulation of glycogen metabolism by covalent modification and allosteric modulators in liver and muscle.
SOM. MK.I.BPM2.2.DM.2.BCHM.1153	Review regulation of hepatic carbohydrate metabolism after a meal and compare it to the fasting state.

Recommended Reading

- **Lippincott's Illustrated Reviews Biochemistry 9th ed.**
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402> Chapter 8: Section Regulation of Glycolysis. Chapter 10: Section Regulation of gluconeogenesis. Chapter 11: Section Glycogenesis and Glycogenolysis Regulation.
- [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- [Gluconeogenesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- [Glycogen Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Carbohydrate metabolism pathways: Fed vs. Fast

Carbohydrate metabolism pathways in the liver depend on the blood glucose concentration



Glycogenesis



High insulin/glucagon ratio



Glycolysis



Glycogenolysis

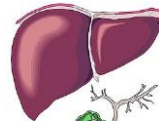


Low insulin/glucagon ratio



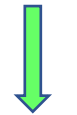
Gluconeogenesis

Regulation of glycolysis and gluconeogenesis in liver



Glucose

Glucokinase



Glucose 6-phosphatase

Glucose 6-P



Fructose 6-P

PFK-1



Fructose 1,6-bisphosphatase

Fructose 1,6-bisP



PEP

PEP Carboxykinase

Pyruvate Kinase



Pyruvate

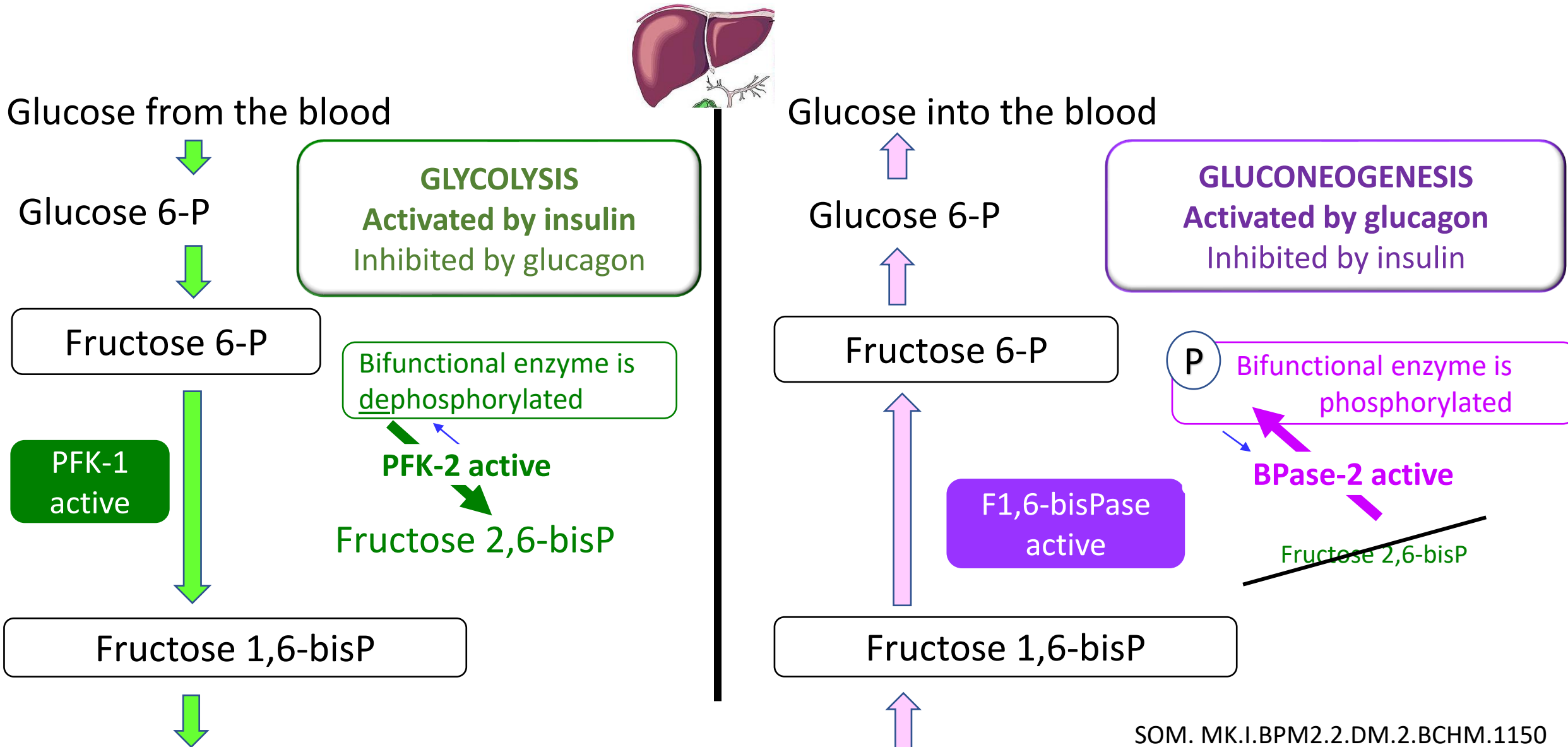
Pyruvate Carboxylase



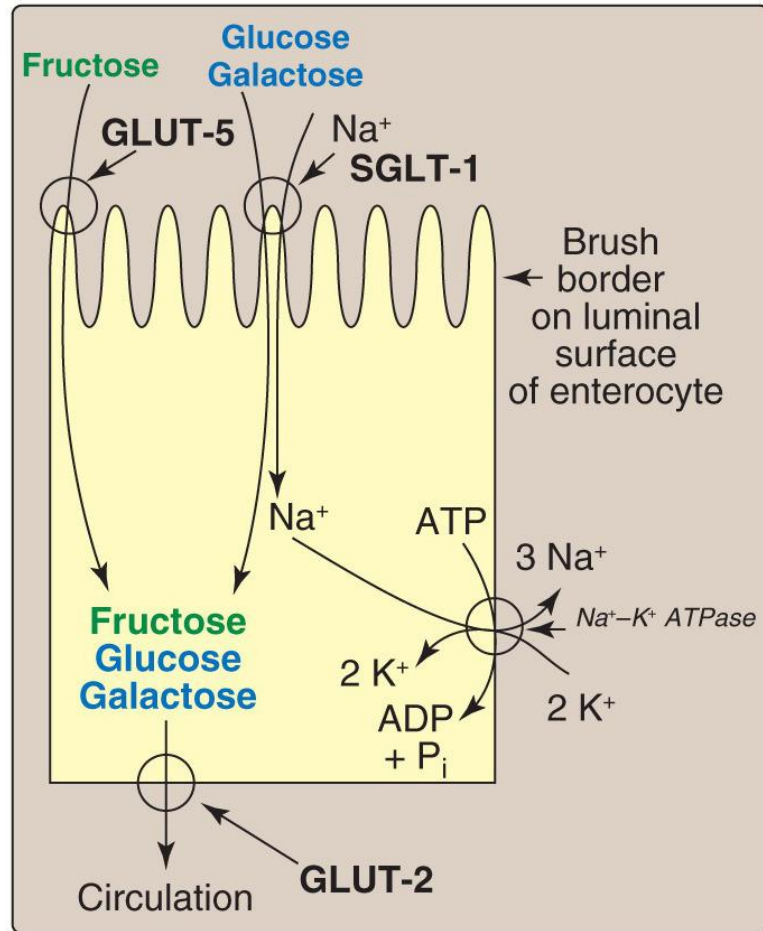
GLYCOLYSIS
Activated by insulin
Inhibited by glucagon

GLUCONEOGENESIS
Activated by glucagon
Inhibited by insulin

Hepatic bifunctional enzyme: Reciprocal regulation of glycolysis and gluconeogenesis

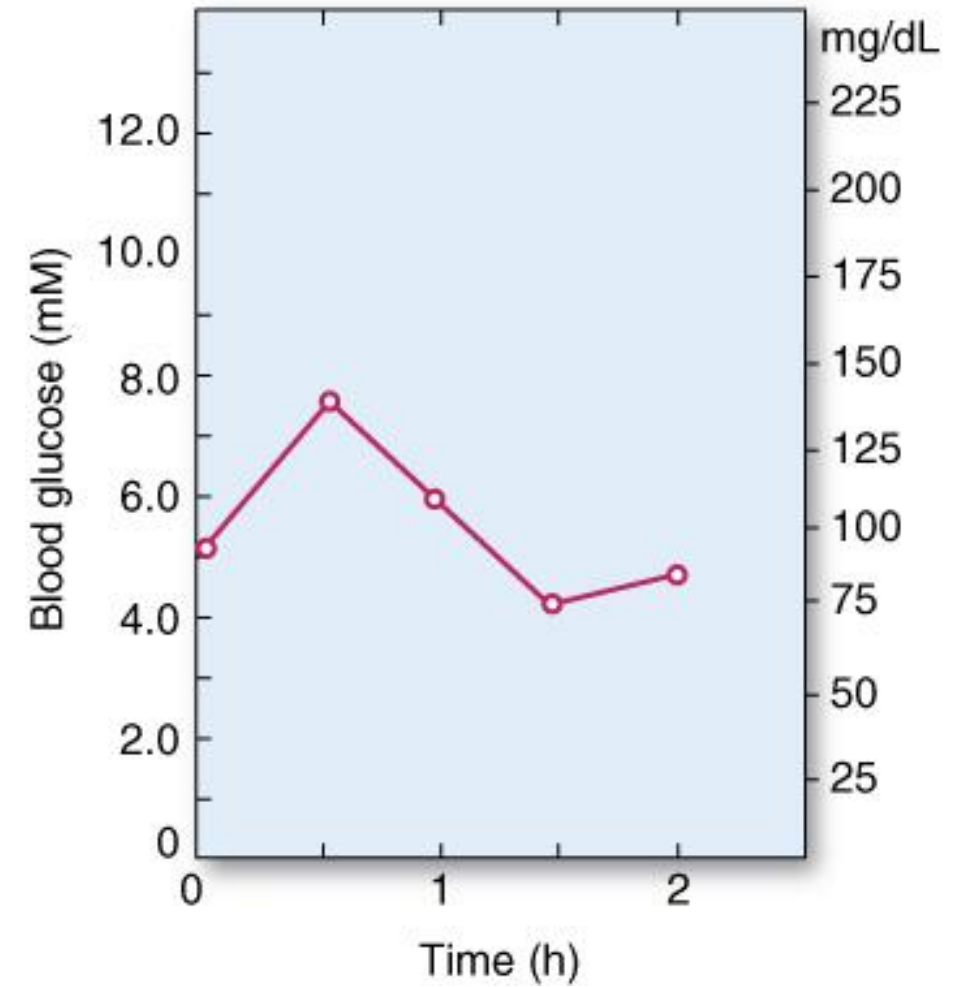


Dietary glucose absorption



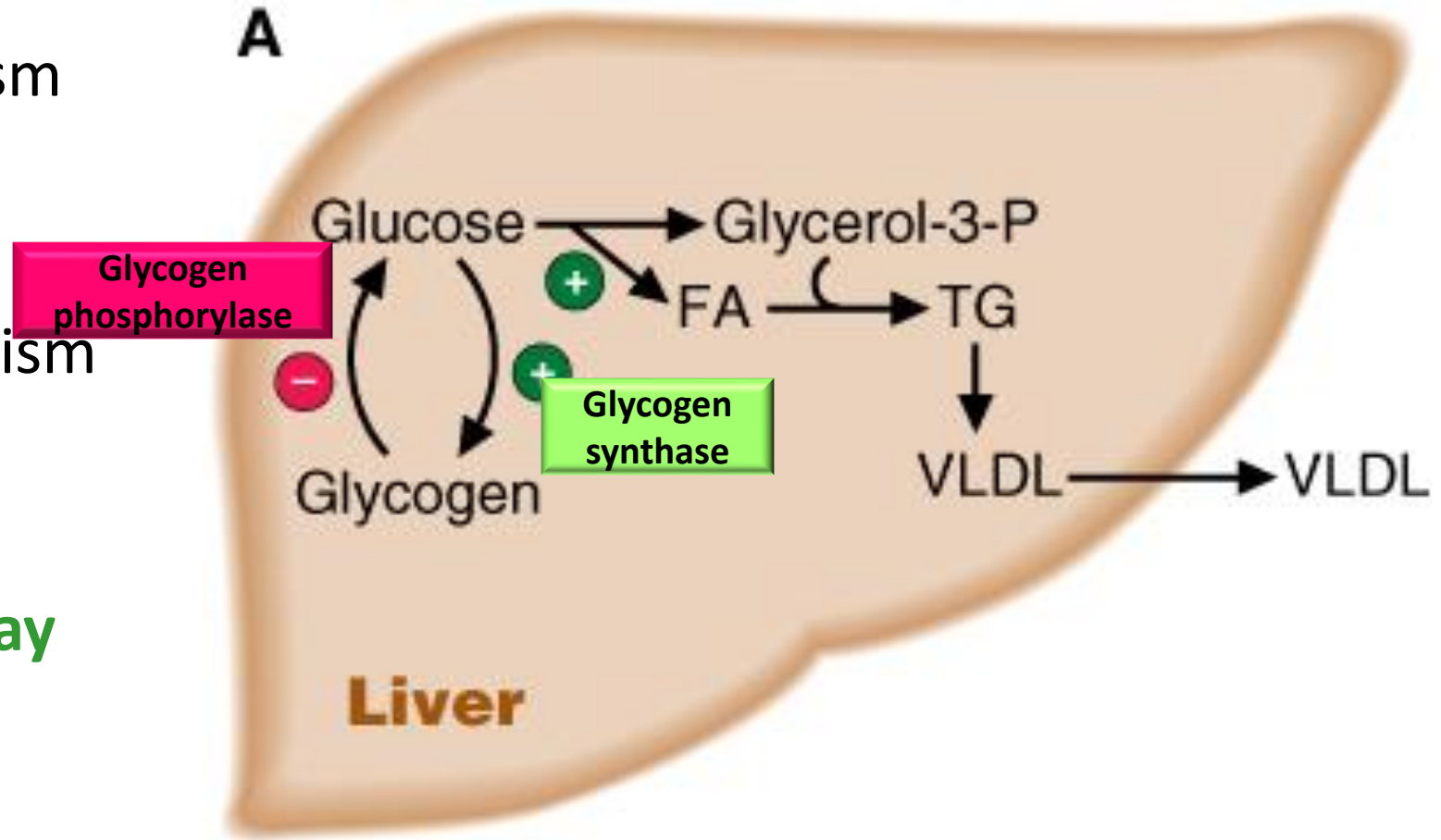
SGLT-1 on luminal side of mucosal cells
GLUT-2 transports glucose into blood

Plasma glucose levels following a meal



Fate of dietary glucose in liver

- Effect on glucose metabolism
 - Glycolysis
 - Gluconeogenesis
- Effect on glycogen metabolism
 - Glycogenesis
 - Glycogenolysis
- Pentose phosphate pathway



Summary: Actions of insulin to reduce blood glucose level

Liver

- **Stimulates glycolysis**
 - PFK-1: Allosteric activation ↑↑Fructose 2,6-bisP
 - Pyruvate kinase (active)
 - Induction of key enzymes (long term)
- **Inhibits gluconeogenesis**
 - Fructose 1,6-bisPase: ↑↑Fructose 2,6-bisP
Allosteric inhibition
- **Stimulates glycogenesis**
 - Active Glycogen synthase (covalent modification)
- **Inhibits glycogenolysis**
 - Inactive Glycogen phosphorylase (covalent modification)
- **Liver uses glucose and stores glucose as glycogen**

Peripheral tissues

- Glucose uptake (↑ GLUT4)
- Activates glycolysis
- Glycogenesis in muscle
- TAG synthesis and storage in adipocytes
- **Peripheral tissues use glucose for energy and storage**

Liver in fed state

Glycogen synthase is activated covalently (dephosphorylated) and allosterically (by glucose 6-P).

Glucose 6-P availability stimulates the pentose phosphate pathway, providing NADPH for fatty acid synthesis.

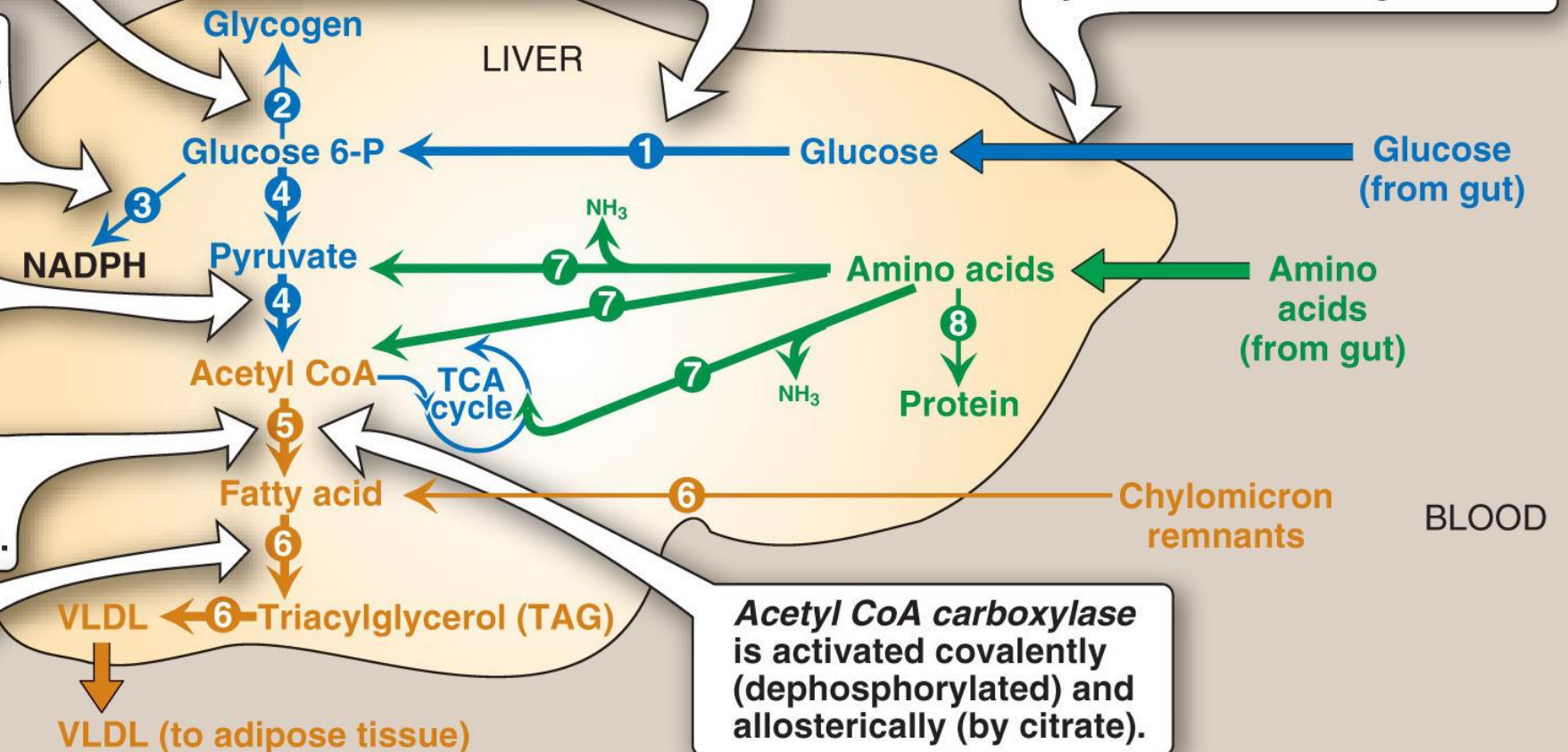
Activation (dephosphorylation) of *pyruvate dehydrogenase* favors acetyl CoA production.

TCA cycle inhibition at *isocitrate dehydrogenase* allows use of acetyl CoA in fatty acid synthesis.

Glycolysis provides the glycerol backbone for TAG synthesis.

The rise in glucose allows phosphorylation by *glucokinase*, which has a high K_m for glucose.

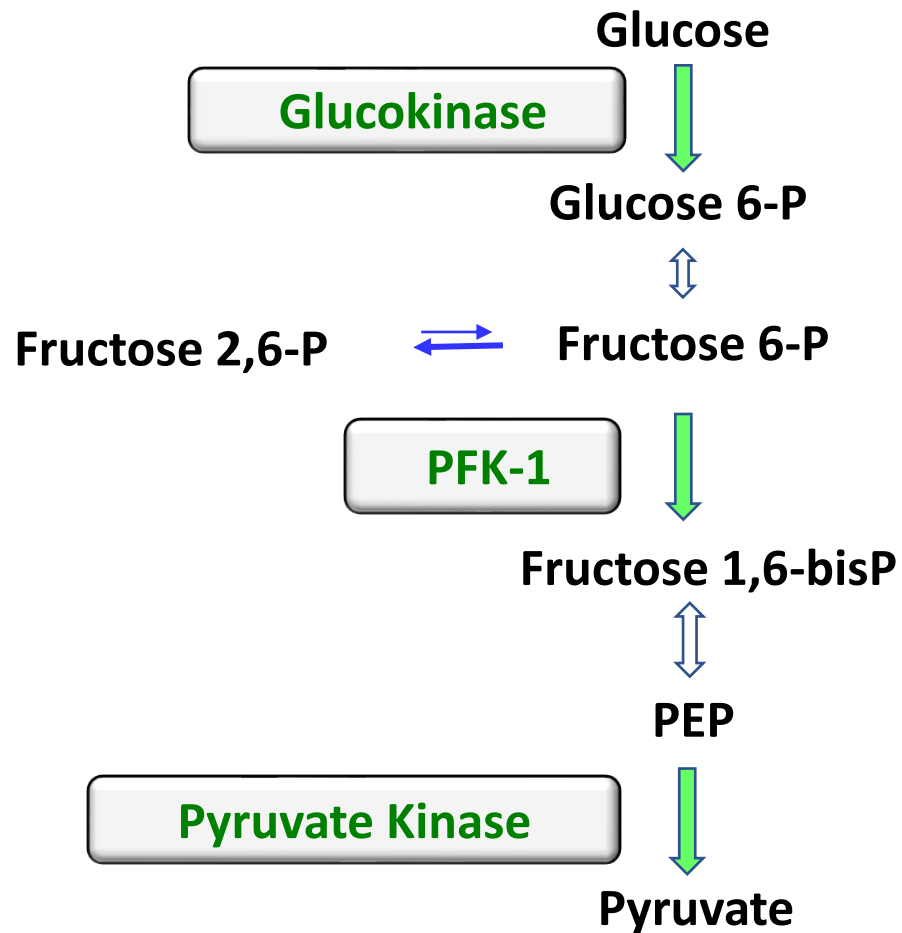
Glucose uptake by insulin-independent GLUT-2 is driven by the rise in blood glucose.



BLOOD

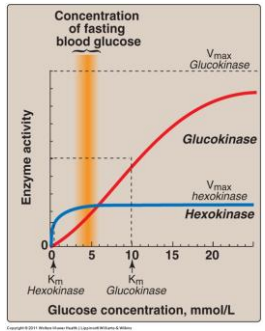
Regulation of Liver Glycolysis:

Regulation of **glucokinase**, **PFK-1** and **hepatic pyruvate kinase**

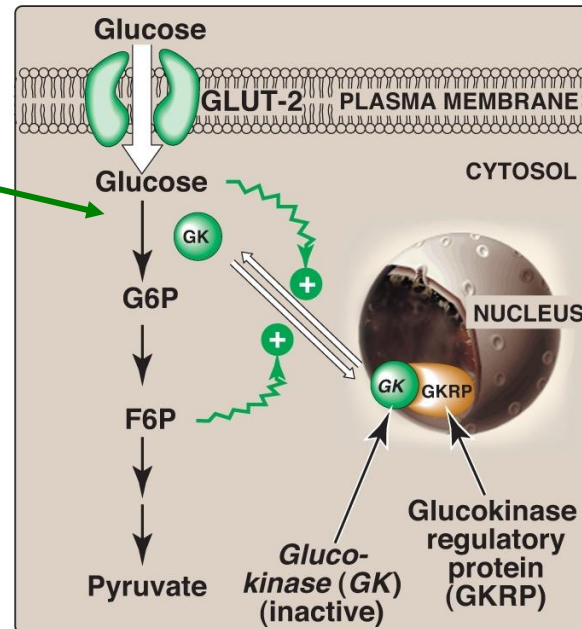


Glucokinase

- First enzyme of glycolysis in **hepatocytes** and **β cells of pancreas**
- Role in **hepatocytes: Reduction** of **high blood glucose** levels after a meal.
- Role in **beta-cells: Recognition** of **high blood glucose** levels and **release of insulin**.
- Defects in glucokinase result in **Monogenic diabetes (MODY-2)**.



At **high blood glucose**,
glucokinase is **active**
in **cytosol**.



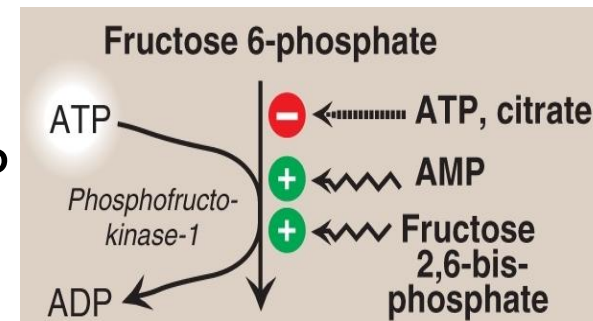
At **low cytosolic glucose**,
glucokinase is sequestered
in nucleus and **inactive**

Regulation of hepatic PFK-1

Insulin (fed state):

- **Dephosphorylates** the bifunctional enzyme
- **Dephosphorylated** bifunctional enzyme **forms** fructose 2,6-bisphosphate → **activates** **allosterically** **PFK-1** (promoting glycolysis) and **inhibits** fructose 1,6-bisphosphatase (suppressing gluconeogenesis).

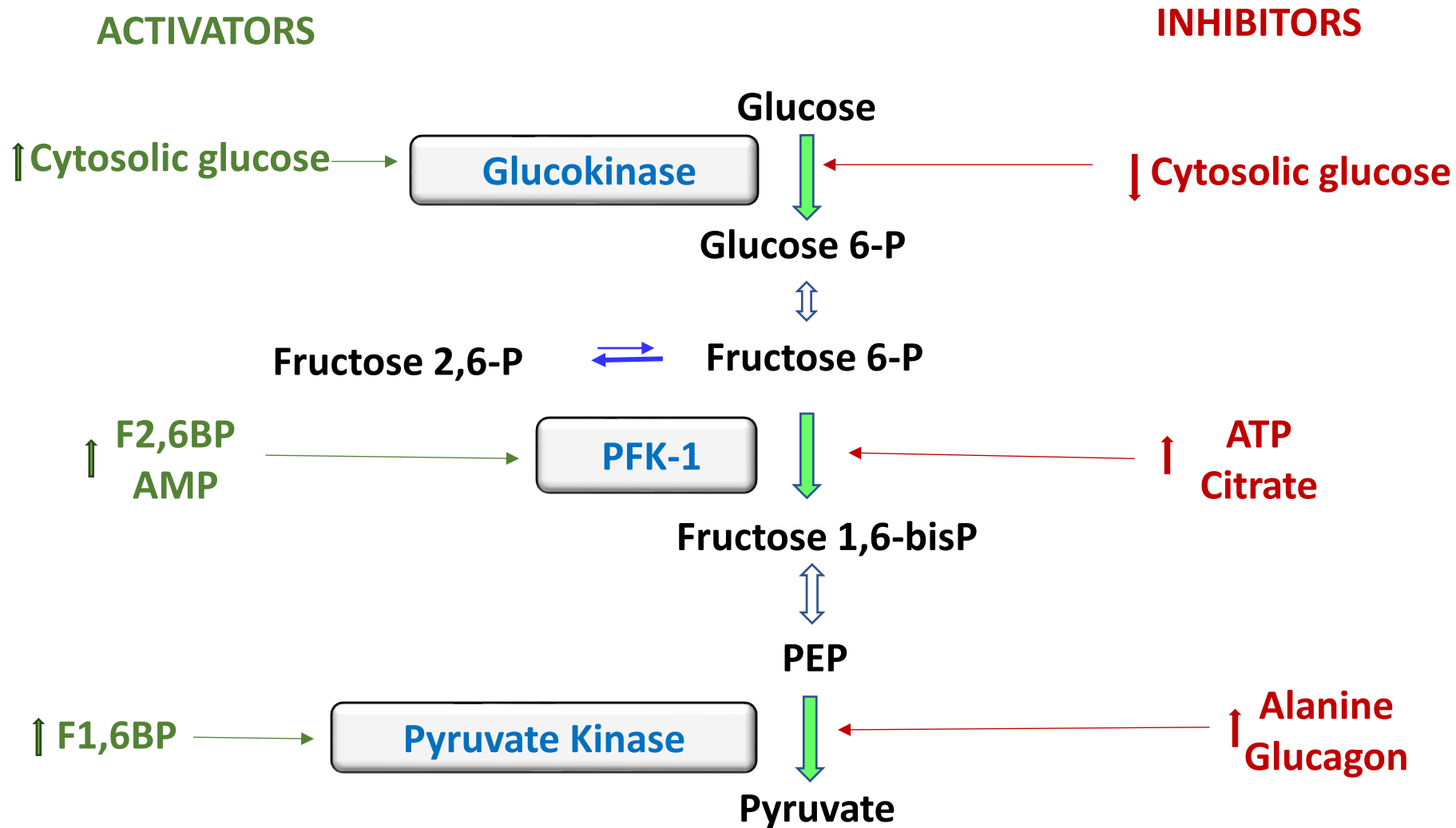
AMP signals low ATP (energy) → **Activates** liver glycolysis to form ATP



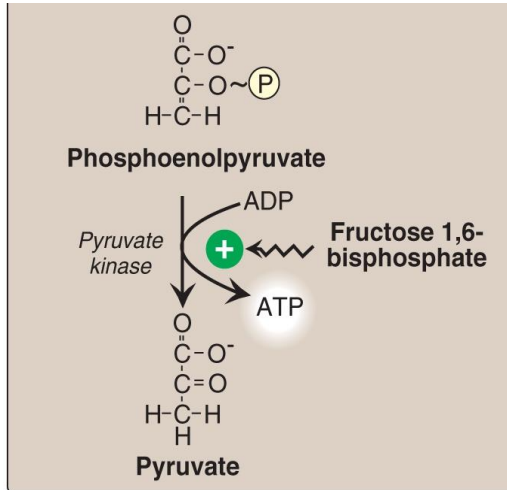
Healthy liver typically has normal ATP levels which inhibit PFK-1.

This inhibition overcome by fructose 2,6-bisP and AMP.

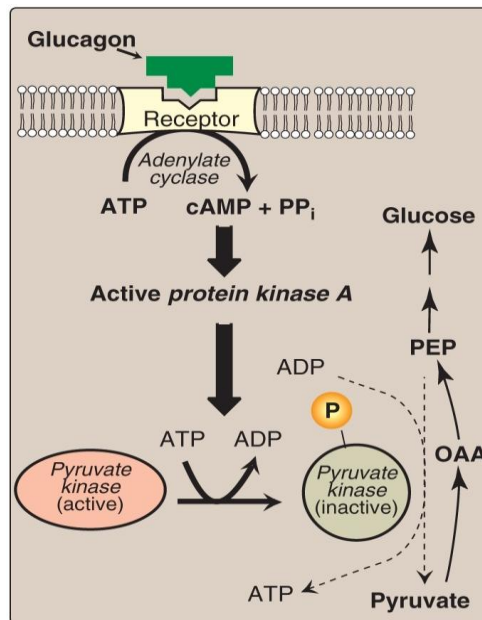
Summary: Regulation of Liver Glycolysis:



Regulation of hepatic pyruvate kinase



High blood glucose $\rightarrow$ insulin $\rightarrow$ Activates glycolysis $\rightarrow$ High **Fructose 1,6-bisP activates** hepatic pyruvate kinase allosterically (feed-forward activation)



Hepatic pyruvate kinase **inhibited** during fasting by:

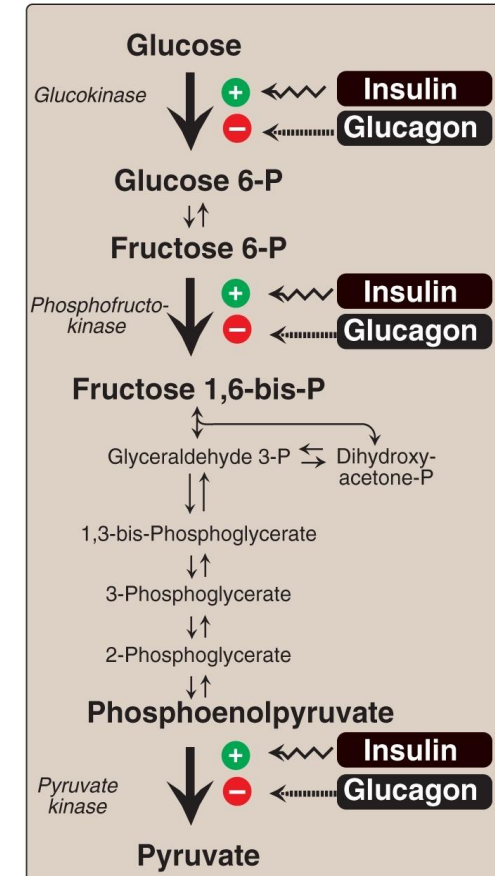
1. **Glucagon** $\rightarrow$ PKA $\rightarrow$ **phosphorylation**
2. **Alanine** (allosterically): When alanine levels are elevated, gluconeogenesis is active; Glycolysis less active

Glycolysis Regulation: Long-term

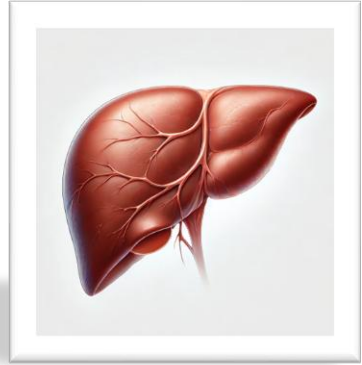
Hepatic glycolysis regulated by transcriptional (long-term) control by insulin or glucagon

Insulin **induces** and glucagon **represses** synthesis of key glycolytic enzymes (Irreversible steps)

Glucokinase
PFK-1
Pyruvate kinase

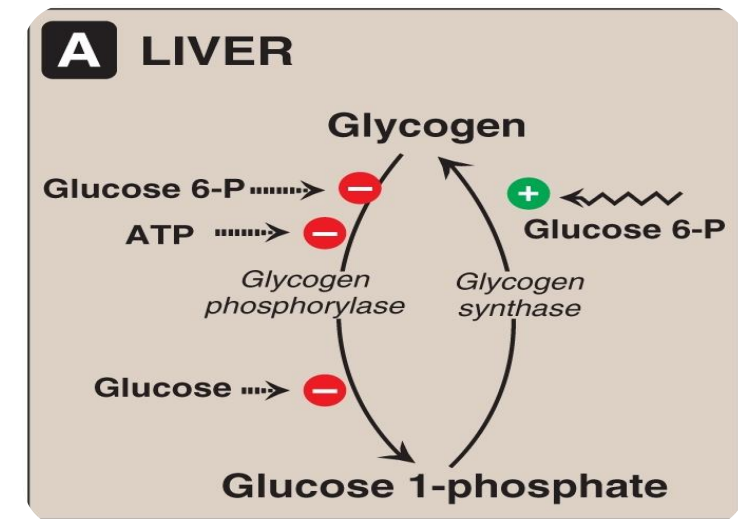


Liver Glycogen Metabolism: Fed State



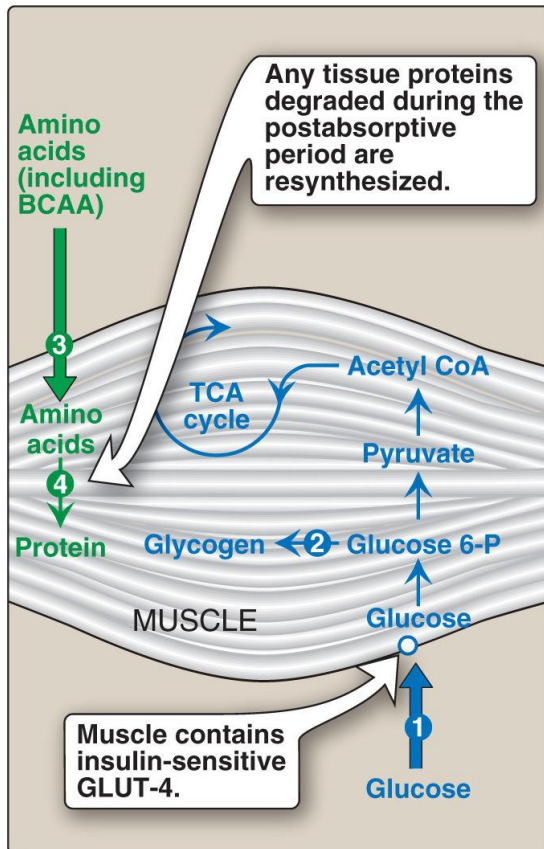
- **Goal: Maintaining blood glucose levels**
- **Allosteric regulation:**
 - **Glycogen synthase:**
 - **Activated** by glucose 6-P
 - **Glycogen phosphorylase:**
 - **Inhibited** by glucose and glucose 6-P
- **Hormonal regulation:**
 - **Glycogen synthase:**
 - **Activated** by insulin (dephosphorylates)

Glycogenesis stores glucose as glycogen
when blood glucose is elevated



Fed State: Fate of Glucose: Adipose Tissue and Muscle

Insulin-induced GLUT-4 uptake of glucose into adipose tissue and skeletal muscle



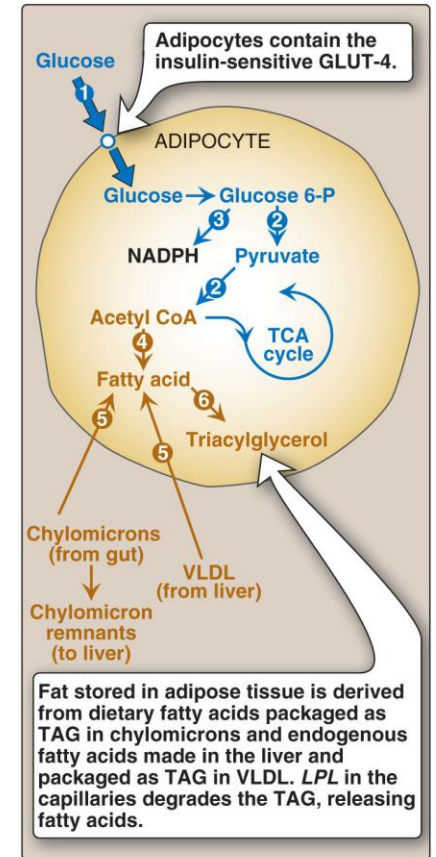
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Glycogenesis in muscle

Glycogen synthase
activated by:

1. Insulin
2. Glucose 6-P

TAG synthesis and storage in adipocytes



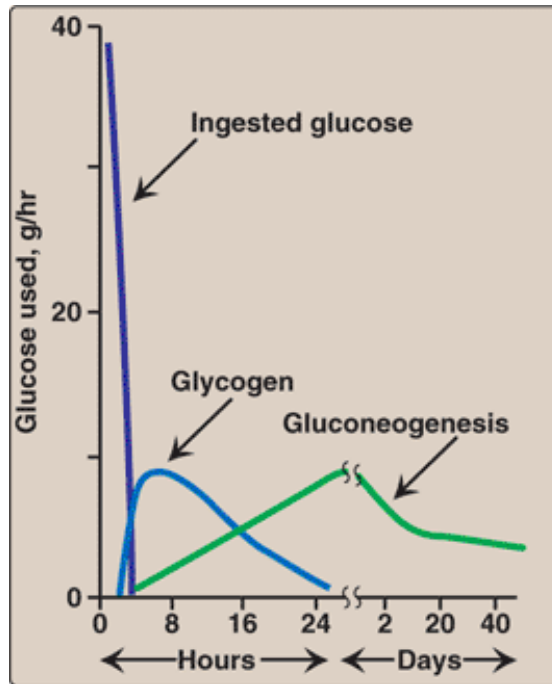
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Fasting blood glucose

Why is it critical to maintain glucose level in fasting state?

- Brain
- RBC

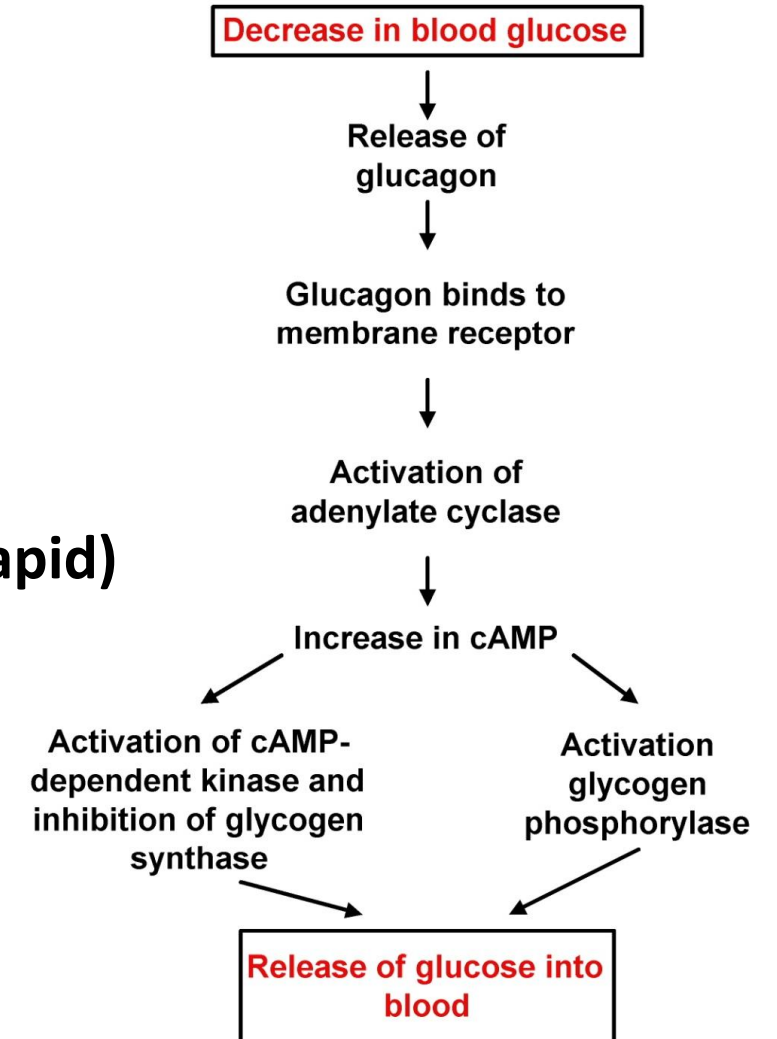
Sources of plasma glucose



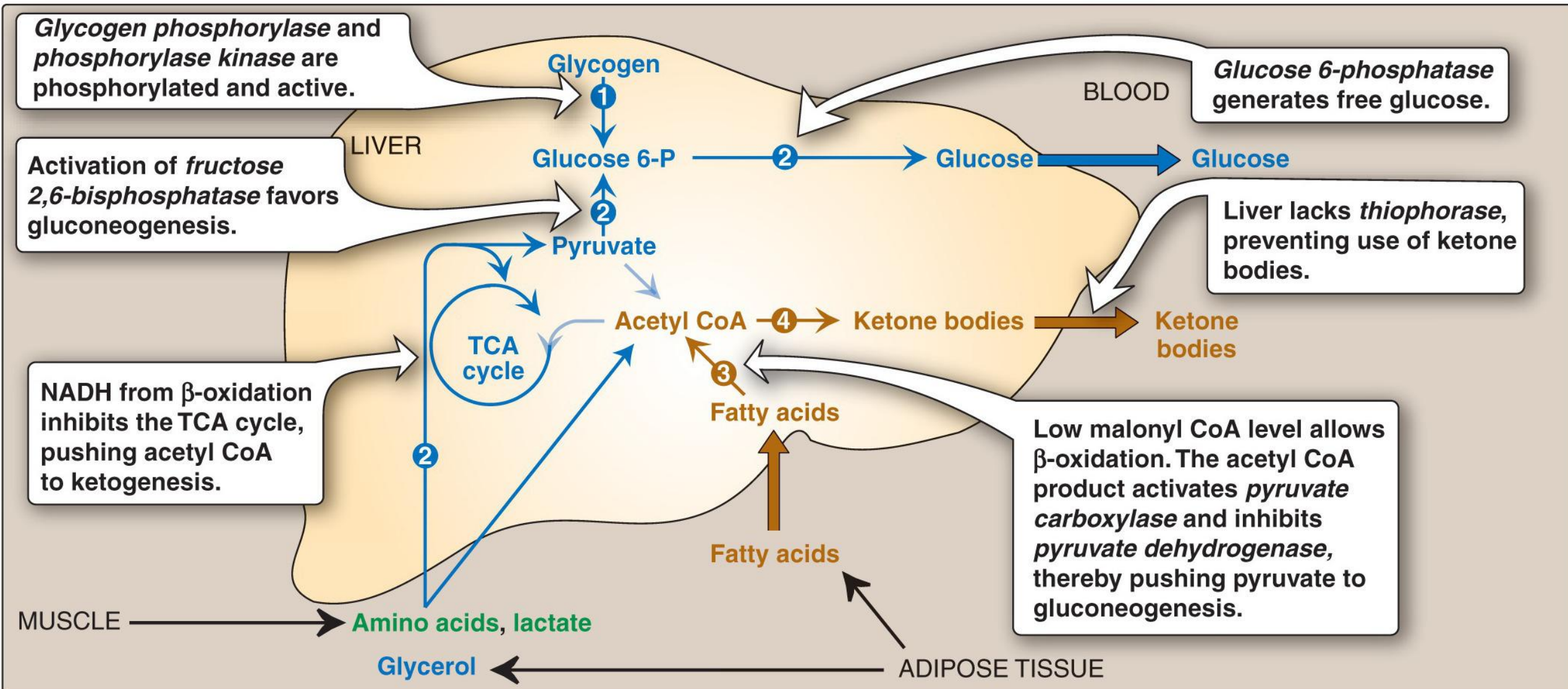
Glucagon effect on liver:

- **Glycogen metabolism**
 - Glycogen phosphorylase (rapid)
 - Glycogen synthase
- Gluconeogenesis
- Glycolysis

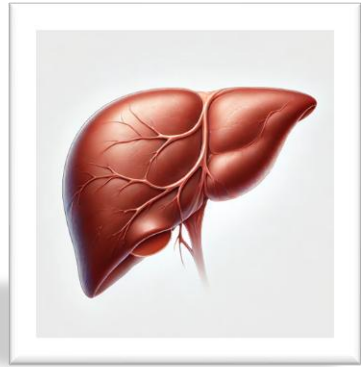
Hormonal changes in fasting



Liver metabolism in fasting state



Liver Glycogen Metabolism in Fasting



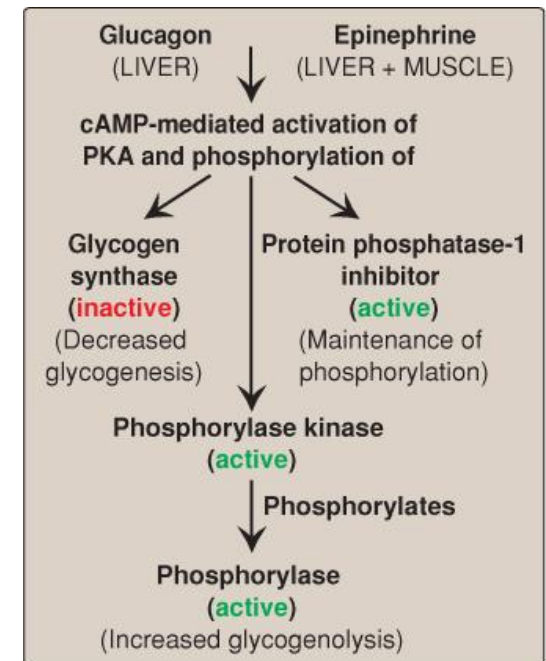
- **Goal: Maintaining blood glucose levels**

Low blood glucose → Glucagon, epinephrine release

Liver glycogenolysis increases blood glucose levels during fasting

- **Hormonal regulation:**
 - **Glycogen phosphorylase kinase:**
 - **Activated** by glucagon
 - **Activated** by epinephrine
 - **Glycogen synthase:**
 - **Inhibited** by glucagon

Activated glycogen phosphorylase kinase → **Activates glycogen phosphorylase**



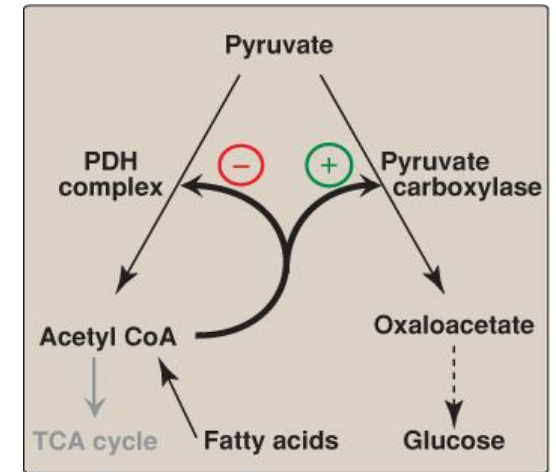
Regulation of Hepatic Gluconeogenesis in Fasting

Glucagon (fasting state):

- **Phosphorylates** the bifunctional enzyme.
- Bifunctional enzyme **degrades** fructose 2,6 bisphosphate → **activates** fructose 1,6-bisphosphatase (promoting gluconeogenesis) and **inhibits** PFK-1 (reducing glycolysis).
- **Phosphorylates** and **inhibits** hepatic pyruvate kinase → Use of PEP for gluconeogenesis.

Transcriptional control: (Long term)

- **Glucagon induces** gluconeogenesis enzymes:
 - **PEP carboxykinase**
 - **Fructose 1,6-bisphosphatase**
 - **Glucose 6-phosphatase**
- **Cortisol induces** PEP carboxykinase and **aminotransferases**.



Fatty acid oxidation



Acetyl CoA

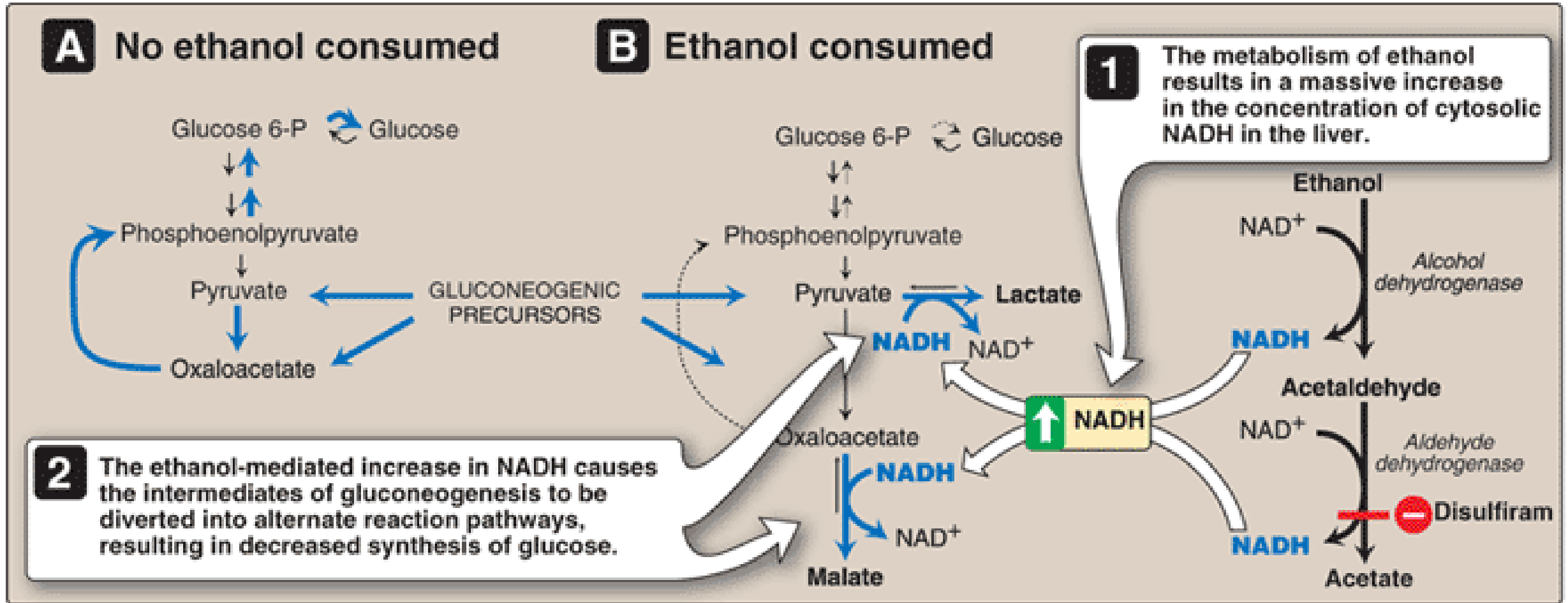


Activates Pyruvate carboxylase

Disorders that present with hypoglycemia

- Acute alcohol intoxication
- Glycogen storage disorders:
 - von Gierke disease, Cori disease
- Hereditary fructose intolerance
- Classical galactosemia
- Liver fatty oxidation defect
- ER module:
 - Insulinoma
 - Addison disease
 - Congenital adrenal hyperplasia (CYP21)
 - Insulin overdose and sulfonylurea overdose

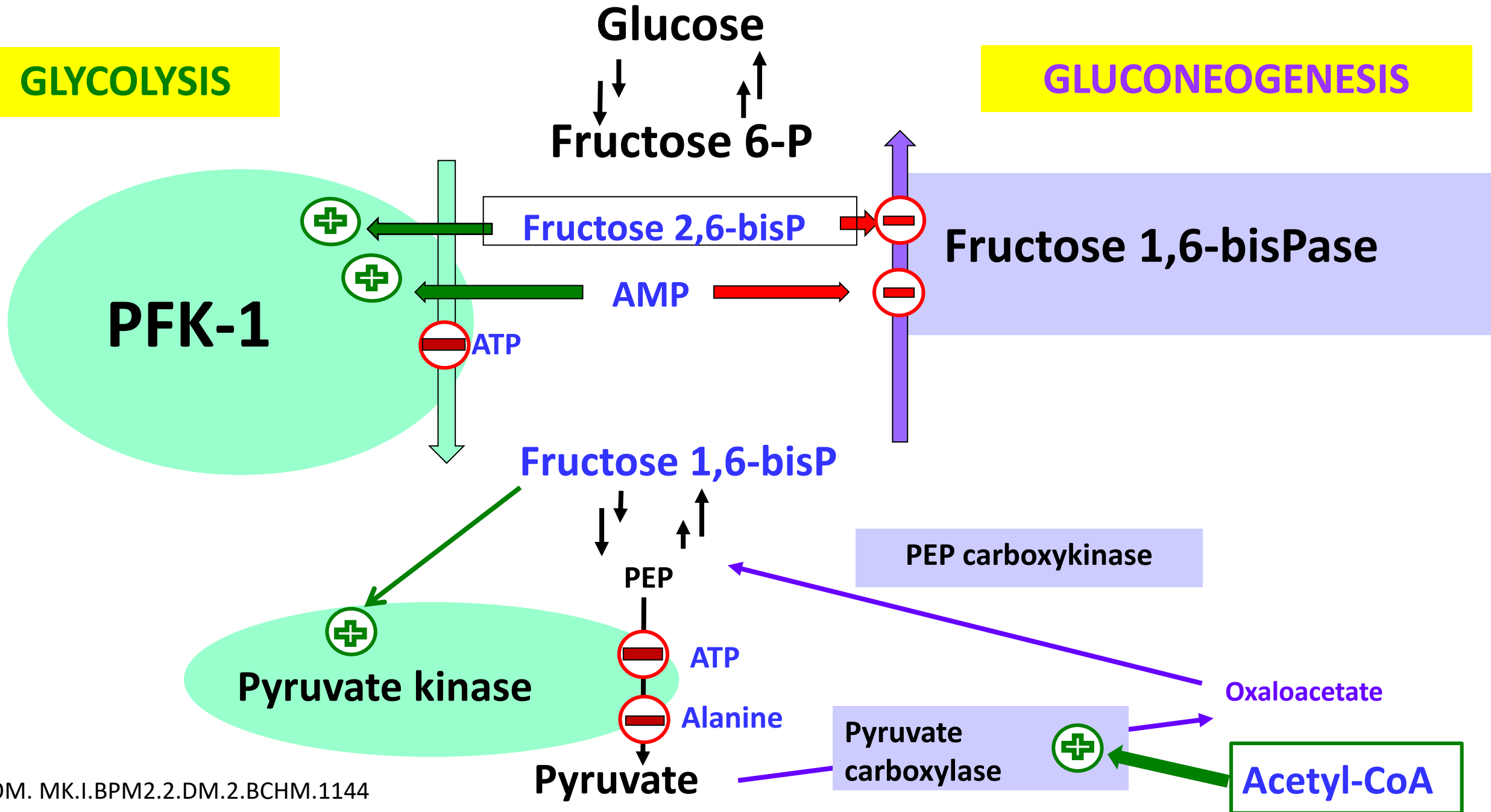
Alcohol Induced Hypoglycemia



Summary of hepatic *allosteric* regulations

GLYCOLYSIS

GLUCONEOGENESIS



Glycogen Synthesis vs. Glycogen Breakdown in Liver and Muscle

Glycogen synthesis:

Glycogenesis **stores** excess glucose as **glycogen** when blood glucose is abundant

Glycogen degradation:

Purpose is **different** in **liver** and **muscle**

Liver glycogenolysis **increases blood** glucose levels

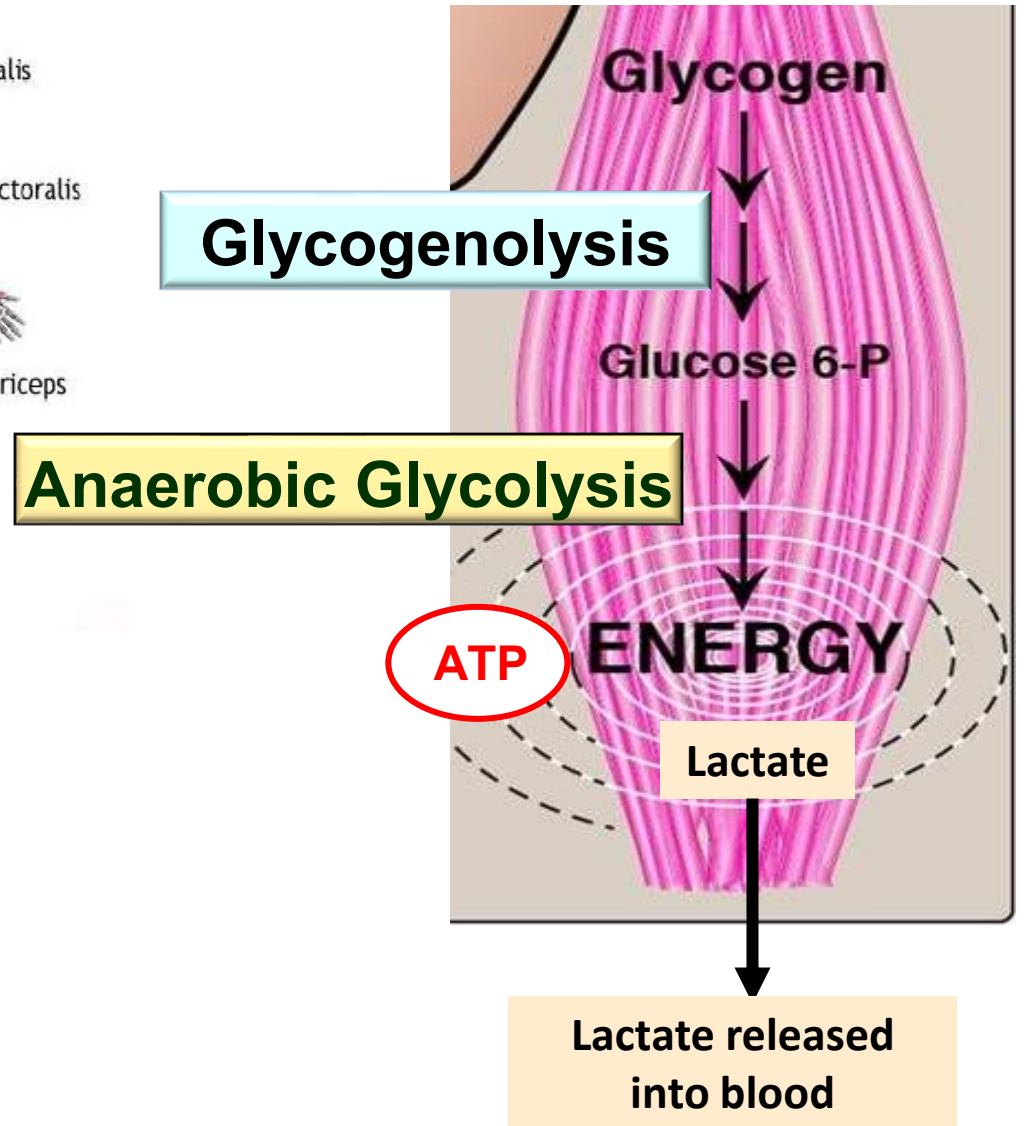
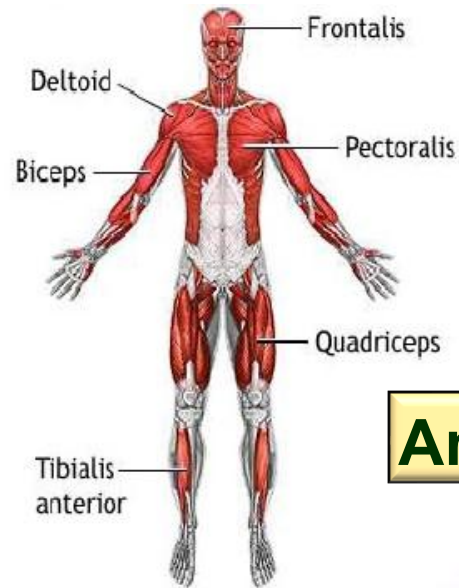
Muscle glycogenolysis generates **energy** for muscle contraction (Lactate is released into blood).

Glycolysis in skeletal muscle is linked to glycogen degradation

Skeletal Muscle:

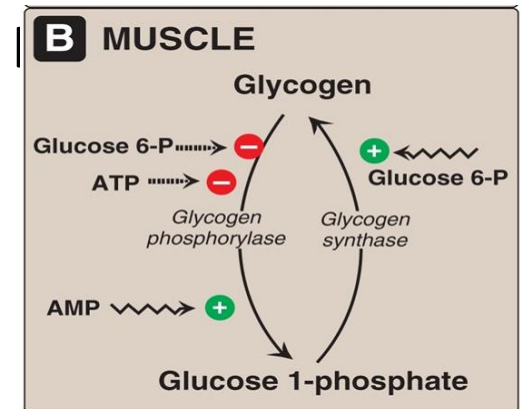
Glycogenolysis and **anaerobic glycolysis** take place during **active muscle contraction**.

Purpose: **ATP formation** followed by release of **lactate** into blood.



Regulation of muscle glycogen degradation

- Muscle contraction → **increases** calcium ions and **AMP** → both **allosterically activate** muscle glycogen degradation
- Epinephrine **optimizes** muscle glycogen degradation (via Protein Kinase A)

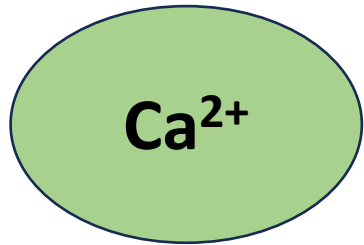


- **Allosteric regulation:**
 - **Calcium** **activates** glycogen phosphorylase kinase
 - **AMP** and glycogen phosphorylase kinase **activate** **glycogen phosphorylase**
 - **ATP** and **glucose 6-P** **inhibit** **glycogen phosphorylase**
- **Hormonal regulation:**
 - **Epinephrine** **activates** glycogen phosphorylase kinase

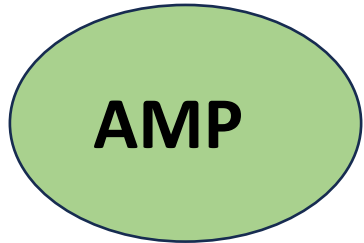
Muscle glycogenolysis: Epinephrine, Calcium and AMP



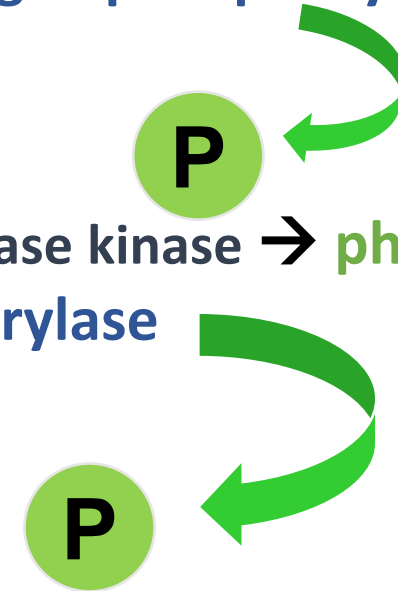
Epinephrine **optimizes** muscle glycogen degradation → **PKA phosphorylates** and **activates** **glycogen phosphorylase kinase**



Calcium activates → **Glycogen phosphorylase kinase** → **phosphorylates**
Glycogen phosphorylase

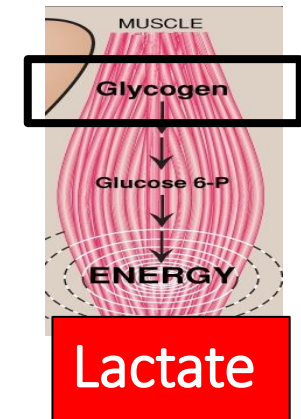
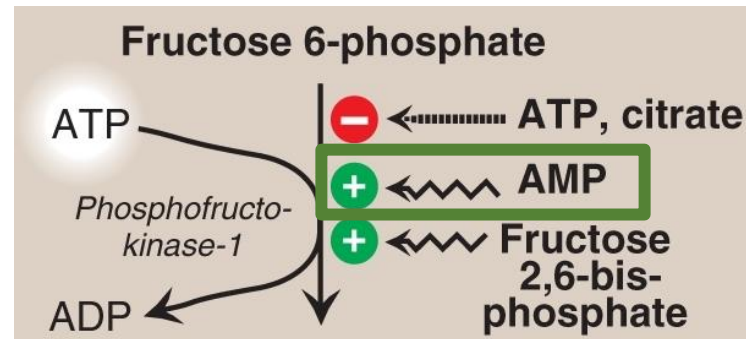


Glycogen phosphorylase
is **activated** by **AMP** and **phosphorylation**



Glycolysis follows glycogen degradation in muscle

AMP activates both muscle glycogen phosphorylase and PFK-1



★ PFK-1 of glycolysis is **activated** by

1. AMP
2. F2,6-bisP

ATP replenished by substrate level phosphorylation in anaerobic glycolysis and lactate is released

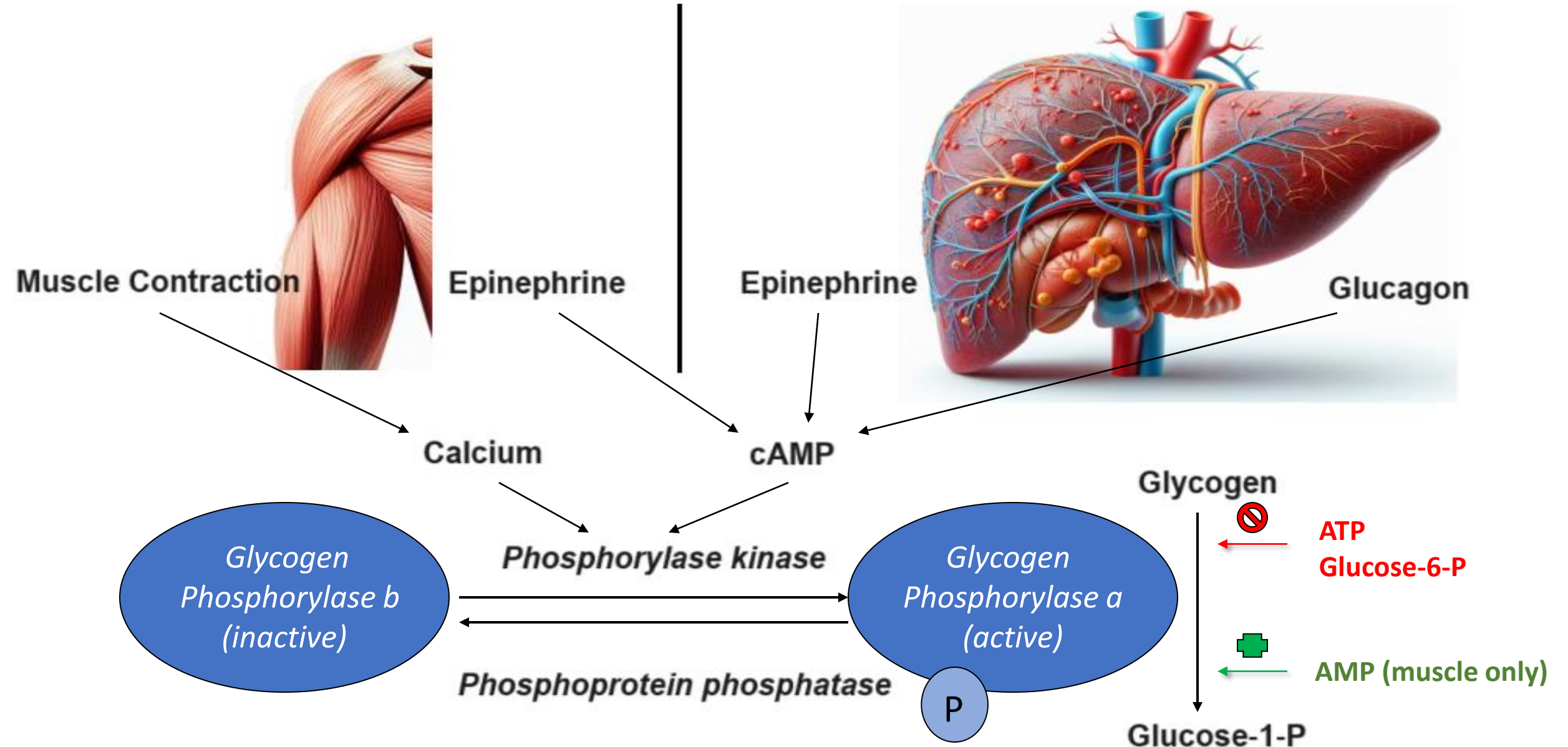
Summary: Muscle energy metabolism during exercise

- **Glycogen phosphorylase** converts glycogen to glucose 1-P.
 - Glucose 1-P → Glucose 6-P for glycolysis → Energy and lactate (anaerobic)
- **PFK-1** (committed step of glycolysis) **activated** by:
 - Muscle contraction → **low ATP**
 - **AMP**

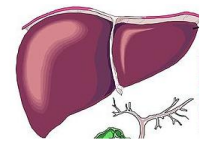
Summary: Regulation of Glycogen Metabolism

- **Glycogen synthase:**
 - **Activated** by dephosphorylation (protein phosphatase-1, **insulin**) and is **inhibited** by **protein kinase A** (cAMP messenger system).
 - Allosterically **activated** by **glucose 6-P**.
- **Glycogen phosphorylase kinase** (used for **regulation** of **glycogen phosphorylase**):
 1. **Activated** by phosphorylation by **protein kinase A** (cAMP).
 2. Allosterically **activated** by **calcium**.
- **Glycogen phosphorylase** (acts on **glycogen** and forms **glucose 1-P**) :
 1. **Activated** by phosphorylation by **glycogen phosphorylase kinase**.
 2. Allosterically **inhibited** by **glucose 6-P** and **ATP** (both **liver** and **muscle**) and by free **glucose** in the **liver**.
 3. In **skeletal muscle** is allosterically **activated** by **AMP** during **muscle contraction**.

Summary: Glycogenolysis regulation: Liver and muscle



Enzyme activities of carbohydrate metabolism in the liver:

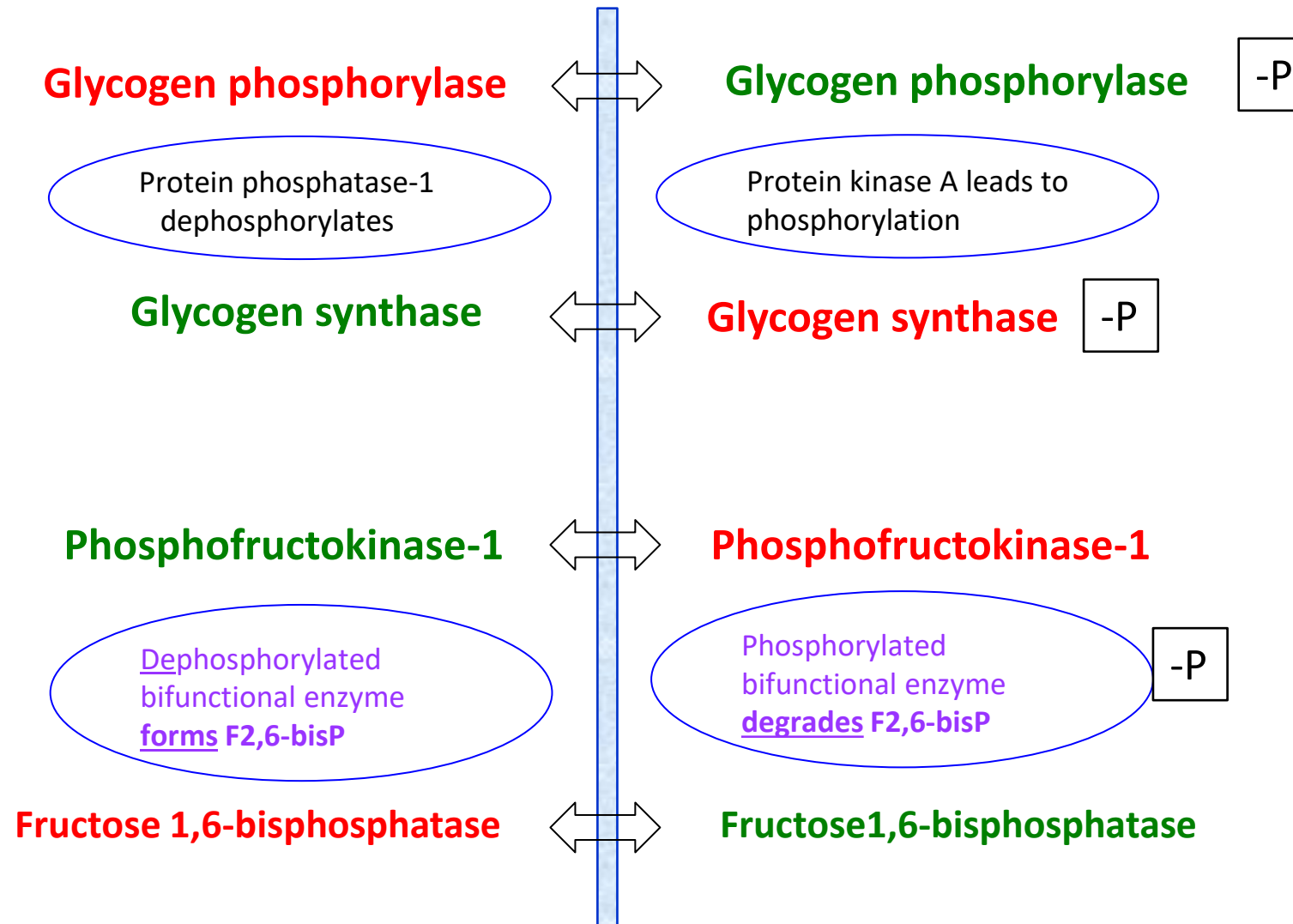


Red indicates low activity and green indicates high activity

After a meal:



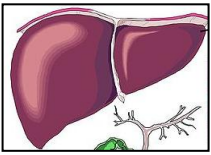
Storage and use of glucose to lower the blood glucose level



During fasting:



Release of glucose into blood to prevent the drop in normal blood glucose



Review: Liver CHO metabolism after a meal



Glycogen synthesis active.

- Glycogen synthase dephosphorylated and **active**.

Glycogen degradation inactive.

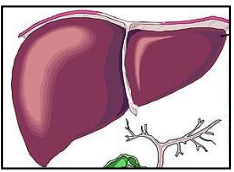
- Glycogen phosphorylase kinase dephosphorylated and **inactive**.
- Glycogen phosphorylase dephosphorylated and **inactive**.

Glycolysis active.

- Glucokinase in **cytosol** (high cytosolic free glucose).
- PFK-1 active: High F2,6-bisP.
- Pyruvate kinase dephosphorylated and **active**: F1,6-bisP (formed by PFK-1) feed-forward activation

Gluconeogenesis inactive.

- Pyruvate carboxylase inactive
- PEP carboxykinase lacks **substrate** in **cytosol**.
- Fructose 1,6-bisphosphatase inhibited by **high F2,6-bisP**.



Review: Liver CHO metabolism during fasting



Glycogen synthesis inactive.

- **Glycogen synthase** phosphorylated and

Glycogen degradation active

- **Glycogen phosphorylase kinase** phosphorylated and **active**.
- **Glycogen phosphorylase** phosphorylated and **active**

Glycolysis inactive:

- **Glucokinase** is in **nucleus**.
- **Bifunctional enzyme** phosphorylated.
 - **PFK-1** is **inactive**: Low levels of F2,6bisP.
- **Pyruvate kinase** phosphorylated and **inactive**; Allosterically **inhibited** by **alanine**

Gluconeogenesis is active:

- **Pyruvate carboxylase** **activated** by **acetyl CoA** (from β -oxidation).
- PDH and TCA cycle are inhibited.
- **PEP carboxykinase** has **oxaloacetate** as **substrate** in **cytosol**.
- **Fructose 1,6-bisphosphatase** not inhibited by F2,6bisP (Low F2,6bisP)

Summary: Allosteric effects of hepatic glycolysis and gluconeogenesis

Effector	Glycolysis	Gluconeogenesis
Indirect: Low cytosolic glucose	Glucokinase is translocated to nucleus.	No allosteric effect
Indirect: High cytosolic glucose	Transport of glucokinase back into cytosol	No allosteric effect
ATP	Inhibits PFK-1 Inhibits pyruvate kinase	No allosteric effect
AMP and fructose 2,6- bisP	Both activate PFK-1	Both inhibit fructose 1,6-bisphosphatase.
Fructose 1,6-bisP	Feed-forward activator of hepatic pyruvate kinase	No allosteric effect
Alanine	Inhibits hepatic pyruvate kinase	No allosteric effector but substrate for gluconeogenesis.
Acetyl CoA	No allosteric effect	Absolute activator of pyruvate carboxylase.

Summary: Hexokinase and Glucokinase

	Hexokinase	Glucokinase
Found in cells and tissues.	In almost all cells	Hepatocytes, glucose-sensing cells like β -cells of pancreas and in neurons of hypothalamus.
Affinity for glucose.	High affinity, low K_m (less than 2 mM) and low V_{max} . Active at low blood glucose concentrations or during hypoglycemia.	Low affinity, high K_m (above normal blood glucose level) and high V_{max} . Very active at high blood glucose levels.
Purpose	Trap glucose in the cell for metabolism. The glucose 6-P cannot leave the cell and used for glycolysis, HMP, or glycogen synthesis.	Liver: traps glucose in cells at high blood glucose levels after a meal. Glucose-sensing cells: "Senses" blood glucose level and release hormones.
Regulation at <u>high</u> blood glucose levels.	Product inhibition by glucose 6-P	No product inhibition by glucose 6-P
Regulation at <u>low</u> blood glucose levels.	Activity dependent only on metabolic needs of cells (product inhibition).	Low cytosolic glucose leads to translocation into the nucleus where the enzyme is inactive.